Uddipt Shankar

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LinkedIn | GitHub | Portfolio

Professional Summary

Final-year CS student at KIIT focused on AI/ML engineering, with hands-on enterprise network security training adding systems-level depth. Builds end-to-end systems using NLP, transformers, and multi-agent LLM architectures, and deploys production-ready AI applications with Docker and scalable backend design.

Technical Skills

Languages	Python, Java
Machine Learning	PyTorch, Scikit-learn, XGBoost, LightGBM
NLP	Hugging Face Transformers, DistilBERT Fine-tuning, NLTK, TF-IDF
LLM & Agents	LangChain, LangGraph, RAG, Tool Calling, Multi-Agent Systems
Backend & APIs	FastAPI, REST APIs, Async Programming
Data, DB & Analysis	PostgreSQL, MySQL, SQLite, Pandas, NumPy, Matplotlib, Seaborn, Power BI
Tools & Deployment	Docker, Render, Streamlit, Git, GitHub, Jupyter Notebook, VS Code
Coursework	Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks
Networking & Security	Cisco ASA, Site-to-Site IPsec VPN, Palo Alto, Packet Tracer, Network Threat Analysis

Experience

Infrastructure Security Intern — HCLTech May 2026 – August 2026

- Built applied understanding of Cisco ASA firewall architecture — security zones, all NAT/PAT types, ACL/object-group design — via hands-on Packet Tracer labs
- Studied Site-to-Site IPsec VPN design: IKEv1/IKEv2 negotiation, ESP vs AH, tunnel mode, Diffie-Hellman groups with PFS, AES-GCM/HMAC-SHA256
- Covered threat-detection fundamentals: DGA-based malicious domain identification, Tor/dark-web architecture, phishing attack lifecycle
- Expanding training into Palo Alto firewalls for cross-vendor security infrastructure exposure

Intern — Sysnet Global Technologies Pvt. Ltd. Oct 2025

Hand Gesture Recognition using OpenCV

- Assisted in building a hand gesture recognition prototype using OpenCV; implemented image preprocessing and tested gesture detection under varied lighting and environmental conditions.
- Gained hands-on exposure to computer vision project workflows, contributing to end-to-end pipeline development used in an industry setting.

Projects

NutriAI — Multi-Agent LLM Nutrition Assistant LangGraph, FastAPI, PostgreSQL, Docker

GitHub | Live API | DockerHub

- Built a **production-grade multi-agent system** (LangGraph) with 3 specialized agents supporting sequential, parallel, and human-in-the-loop workflows.
- Designed a 5-layer nutrition pipeline (cache → DB → fuzzy match → dual-LLM → fallback), achieving **∼99% latency reduction** on repeat queries (26.9ms → 0.2ms, measured) by routing requests away from expensive LLM calls.
- Observed ∼5.4s average latency for novel dishes via dual-LLM inference, with results cached for subsequent <1ms retrieval.
- Reduced hallucination risk using **dual-LLM ensemble** (Qwen + Llama) with divergence detection; observed ∼18.8% average disagreement across novel queries, indicating mitigation of single-model bias.
- Developed a FastAPI backend (25+ REST endpoints) with OCR-based menu ingestion, meal logging, and real-time streaming; async architecture supports 50+ concurrent requests.
- Engineered a context-aware chatbot with tool-calling and **3-layer memory** (short-term state, PostgreSQL persistence, ChromaDB RAG over 500+ dishes).
- Containerized using Docker multi-stage builds and deployed to Docker Hub/Render for one-command deployment.

Ecommerce Review Sentiment Classification ML, NLP, Transformers

GitHub | Live Demo

- Built an end-to-end **sentiment classification system** on ∼100K Amazon reviews; benchmarked Logistic Regression, SVM, Naive Bayes, XGBoost, LightGBM against DistilBERT, optimizing macro-F1 under class imbalance.
- Improved neutral class handling via class-weighted loss and dataset scaling (50K → 100K), achieving ∼90% accuracy and ∼0.69 macro-F1; deployed as a Streamlit app with batch inference, confidence scoring, and real-time visualization.

Inventory Management System Python, MySQL, Streamlit

GitHub

- Designed a **normalized MySQL schema** with 5+ interconnected tables and ACID-compliant stored procedures; built a Streamlit interface for CRUD, real-time tracking, and complex analytical query execution for non technical users.

Additional: Low Level System Design Case Studies | Power BI Dashboard Projects

Education

Kalinga Institute of Industrial Technology (KIIT) Bhubaneswar, India

B.Tech in Computer Science and System Engineering Expected 2027 CGPA: 8.34

Certifications

IBM Machine Learning Professional Certificate | Google Advanced Data Analytics | Google Operating Systems | Meta Database Management | NVIDIA Computer Networking